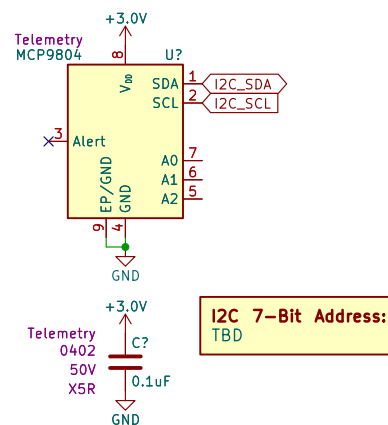
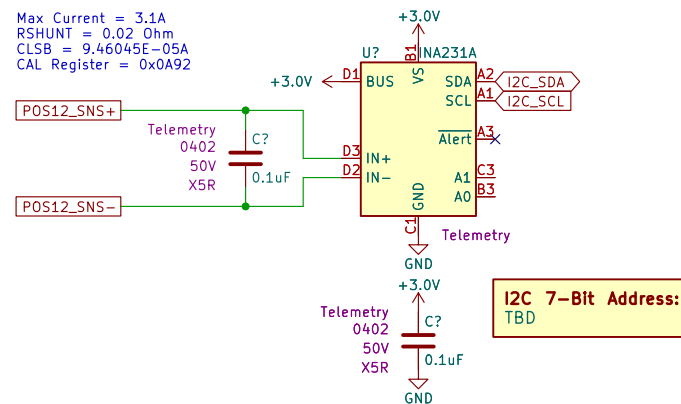


|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |                                                                                                                                                                                 |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 | 5 |                                                                                                                                                                                 |
| A | <div>12V Power Input</div> <div></div> <div>File: POS12_power_input.kicad_sch</div> <div>12V Telemetry</div> <div></div> <div>File: POS12_telemetry.kicad_sch</div> <div>3.3V Power Supply</div> <div></div> <div>File: POS3P3_power_supply.kicad_sch</div> <div>3.3V Telemetry</div> <div></div> <div>File: POS3P3_telemetry.kicad_sch</div> <div>LED Power Supply</div> <div></div> <div>File: led_power_supply.kicad_sch</div> <div>LED Telemetry</div> <div></div> <div>File: led_telemetry.kicad_sch</div> <div>dsPIC33 Programming</div> <div></div> <div>File: dspic33_programming.kicad_sch</div> <div>dsPIC33 Power</div> <div></div> <div>File: dspic33_power.kicad_sch</div> <div>dsPIC33 GPIO</div> <div></div> <div>File: dspic33_gpio.kicad_sch</div> <div>USB UART Bridge</div> <div></div> <div>File: usb_uart_bridge.kicad_sch</div> |   | <div>Platform ETC</div> <div></div> <div>File: platform_etc.kicad_sch</div> <div>PGOOD LEDs</div> <div></div> <div>File: pgood_leds.kicad_sch</div> <div>Status LEDs</div> <div></div> <div>File: status_leds.kicad_sch</div> <div>Audio Input</div> <div></div> <div>File: audio_input.kicad_sch</div> <div>Left RCA Signal Conditioning</div> <div></div> <div>File: left_rca_signal_conditioning.kicad_sch</div> <div>Right RCA Signal Conditioning</div> <div></div> <div>File: right_rca_signal_conditioning.kicad_sch</div> <div>Left Speaker Signal Conditioning</div> <div></div> <div>File: left_speaker_signal_conditoning.kicad_sch</div> <div>Right Speaker Signal Conditoning</div> <div></div> <div>File: right_speaker_signalLconditoning.kicad_sch</div> <div>Programmable Gain Amplifiers</div> <div></div> <div>File: programmable_gain_amplifiers.kicad_sch</div> <div>Power Button</div> <div></div> <div>File: power_button.kicad_sch</div> |   |   |                                                                                                                                                                                 |
| B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |                                                                                                                                                                                 |
| C |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |                                                                                                                                                                                 |
| D |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   | <div></div> <div>Sheet: /<br/>File: Audio_Visualiser_DSP.kicad_sch</div> <div><b>Title:</b></div> <div>Size: ADate: 2023-12-25Rev:</div> <div>KiCad E.D.A. 10.0.3Id: 1/21</div> |
|   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4 | 5 |                                                                                                                                                                                 |





Sheet: /12V Telemetry/  
 File: POS12\_telemetry.kicad\_sch

**Title:**

Size: A Date: 2023-12-25  
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 Id: 3/21

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| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /3.3V Power Supply/  
File: POS3P3\_power\_supply.kicad\_sch

Title:

Size: A

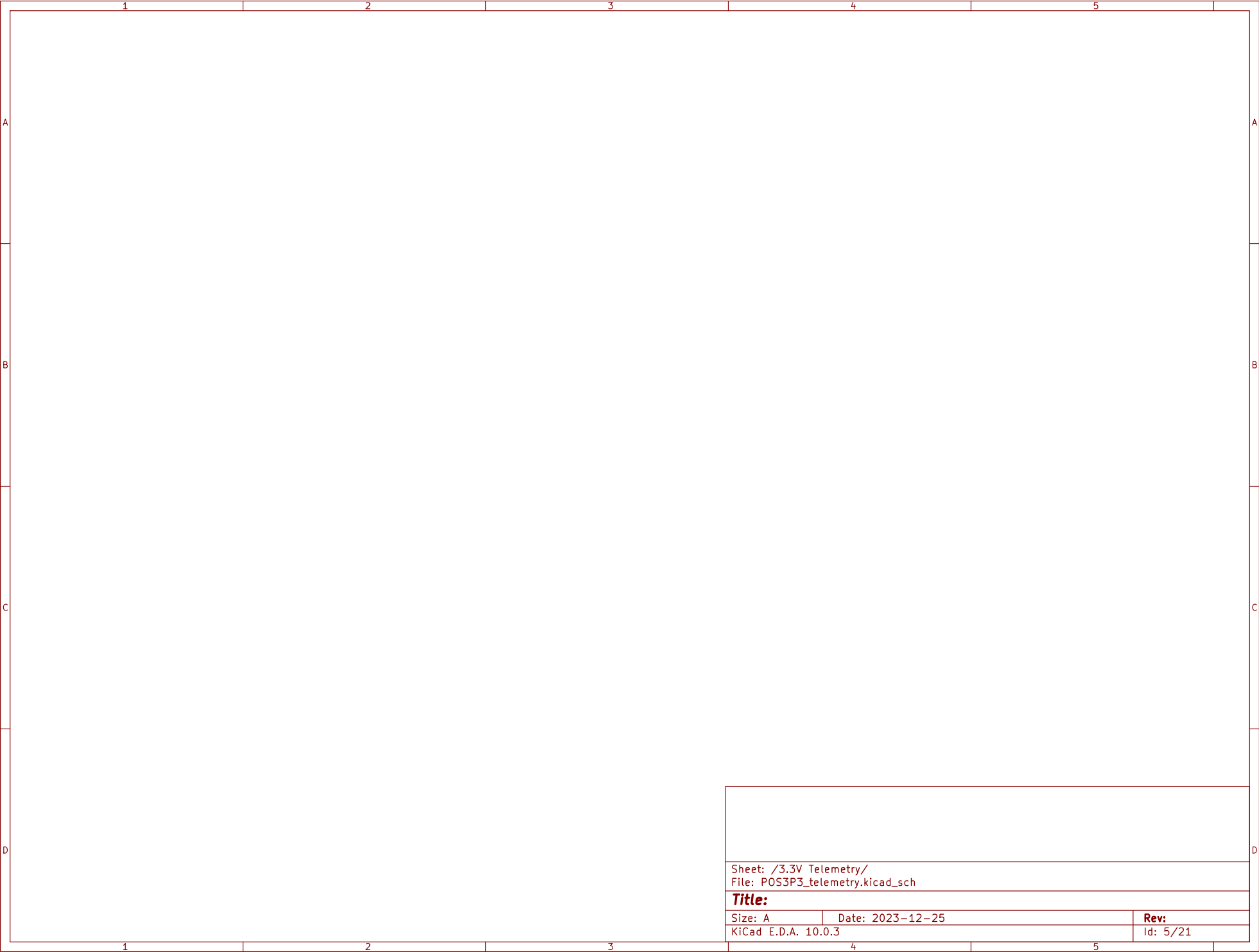
Date: 2023-12-25

Rev:

KiCad E.D.A. 10.0.3

Id: 4/21

|                                                                   |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
|-------------------------------------------------------------------|--|--|--|--|------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|------|--|--|--|--|
| Sheet: /3.3V Power Supply/<br>File: POS3P3_power_supply.kicad_sch |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
| Title:                                                            |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
| Size: A                                                           |  |  |  |  | Date: 2023-12-25 |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Rev: |  |  |  |  |
| KiCad E.D.A. 10.0.3                                               |  |  |  |  | Id: 4/21         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |



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| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /LED Power Supply/  
File: led\_power\_supply.kicad\_sch

Title:

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KiCad E.D.A. 10.0.3Id: 6/21

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|---------------------------------------------------------------|------------------|------------------|
| Sheet: /LED Power Supply/<br>File: led_power_supply.kicad_sch |                  |                  |
| Title:                                                        |                  |                  |
| Size: A                                                       | Date: 2023-12-25 |                  |
| KiCad E.D.A. 10.0.3                                           |                  | Rev:<br>Id: 6/21 |

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| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |
| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /LED Telemetry/  
File: led\_telemetry.kicad\_sch

Title:

Size: A

Date: 2023-12-25

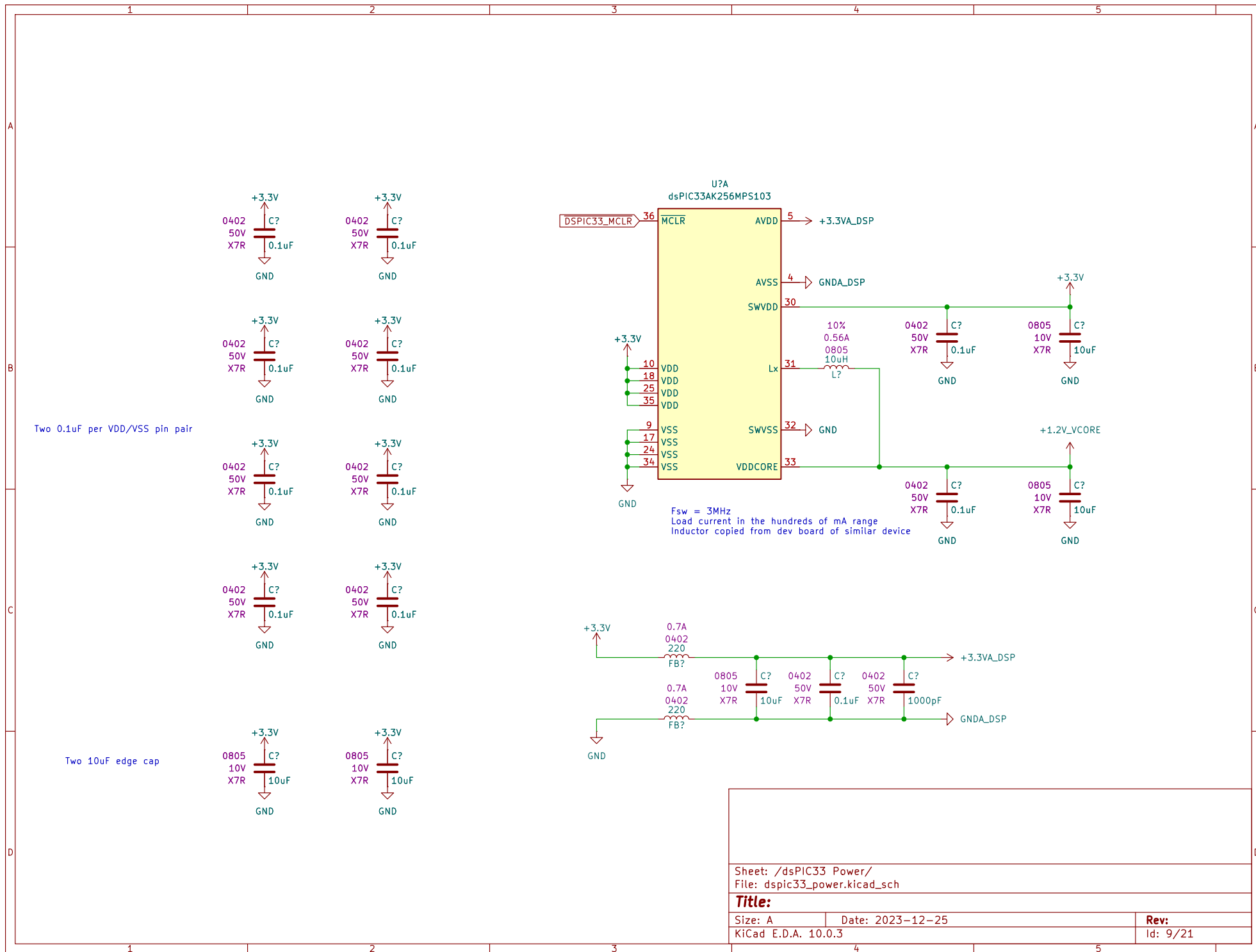
Rev:

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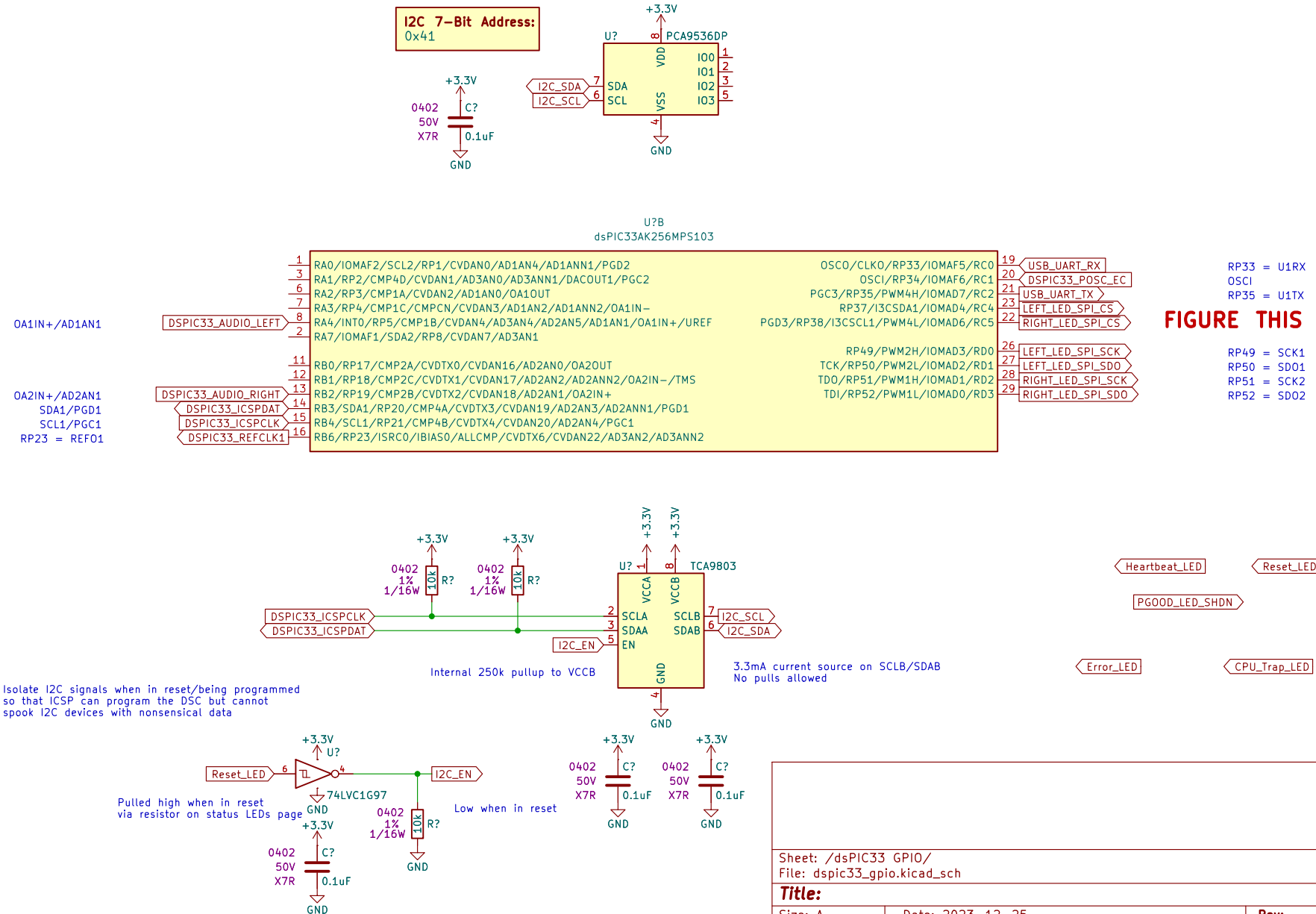
Id: 7/21

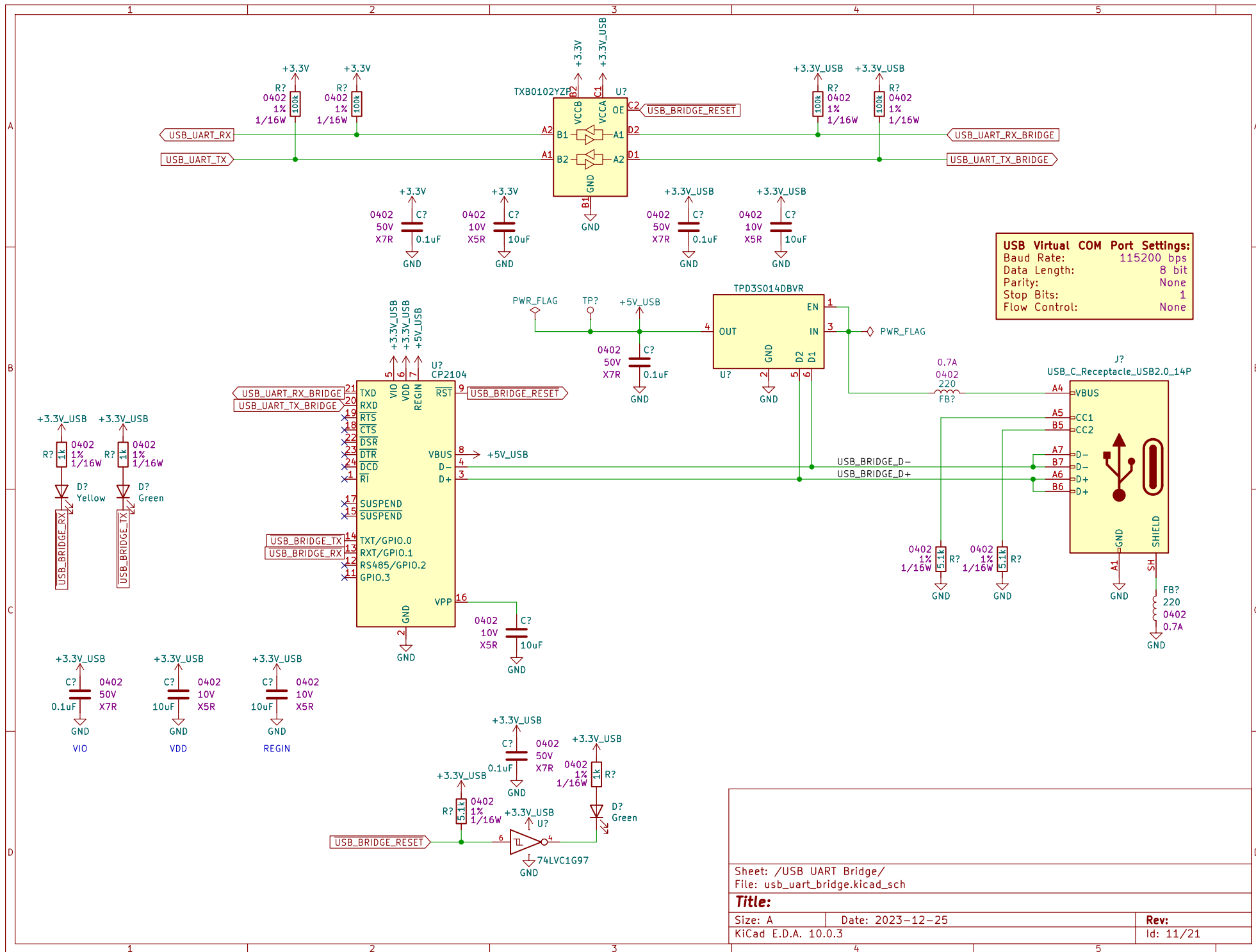
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| Sheet: /LED Telemetry/        |                  |          |
| File: led_telemetry.kicad_sch |                  |          |
| <b>Title:</b>                 |                  |          |
| Size: A                       | Date: 2023-12-25 | Rev:     |
| KiCad E.D.A. 10.0.3           |                  | Id: 7/21 |

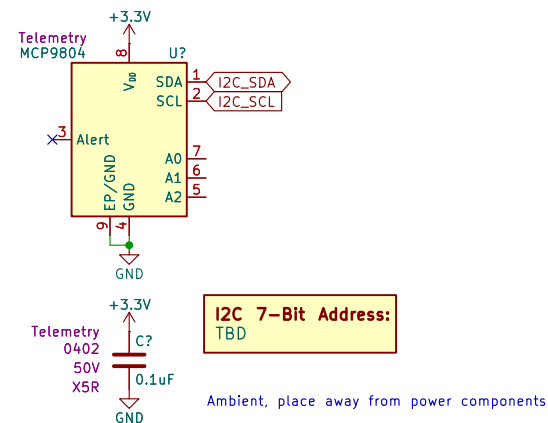
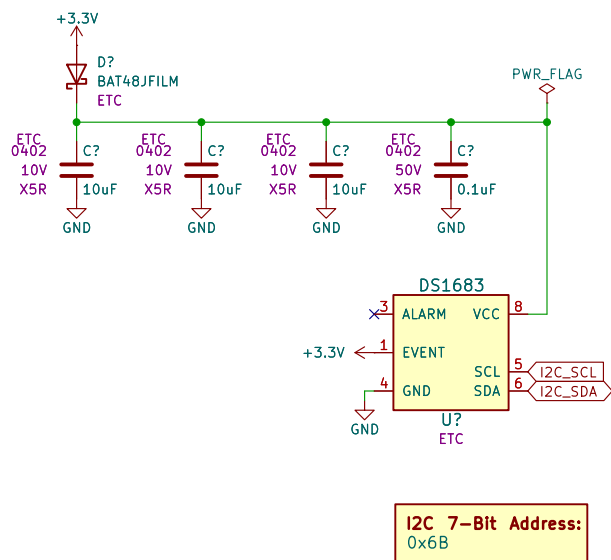




# ADD HARDSTRAPS FOR CONFIG STUFF





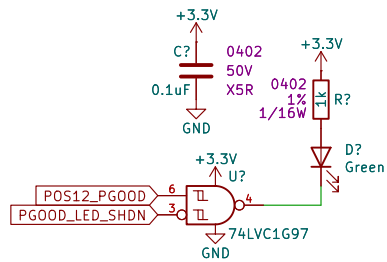
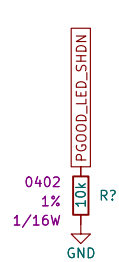


Sheet: /Platform ETC/  
File: platform\_etc.kicad\_sch

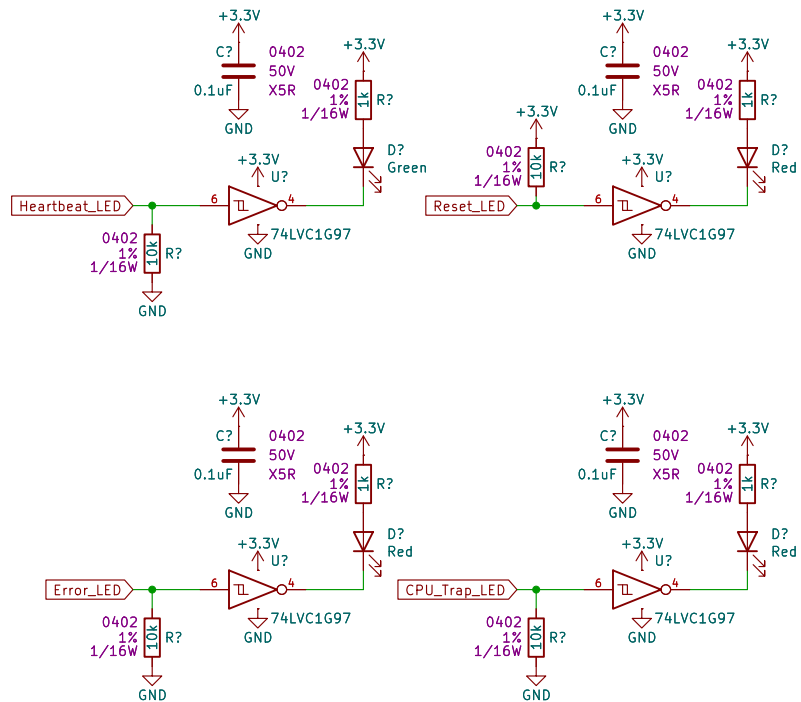
**Title:**

Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

**Rev:**  
Id: 12/21



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|---------------------------------------------------|------------------|-----------|
| Sheet: /PGOOD LEDs/<br>File: pgood_leds.kicad_sch |                  |           |
| <b>Title:</b>                                     |                  |           |
| Size: A                                           | Date: 2023-12-25 | Rev:      |
| KiCad E.D.A. 10.0.3                               |                  | Id: 13/21 |



Sheet: /Status LEDs/  
File: status\_leds.kicad\_sch

**Title:**

Size: A Date: 2023-12-25  
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|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|--|--|--|--|---|
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |
| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /Left RCA Signal Conditioning/  
File: left\_rca\_signal\_conditioning.kicad\_sch

Title:

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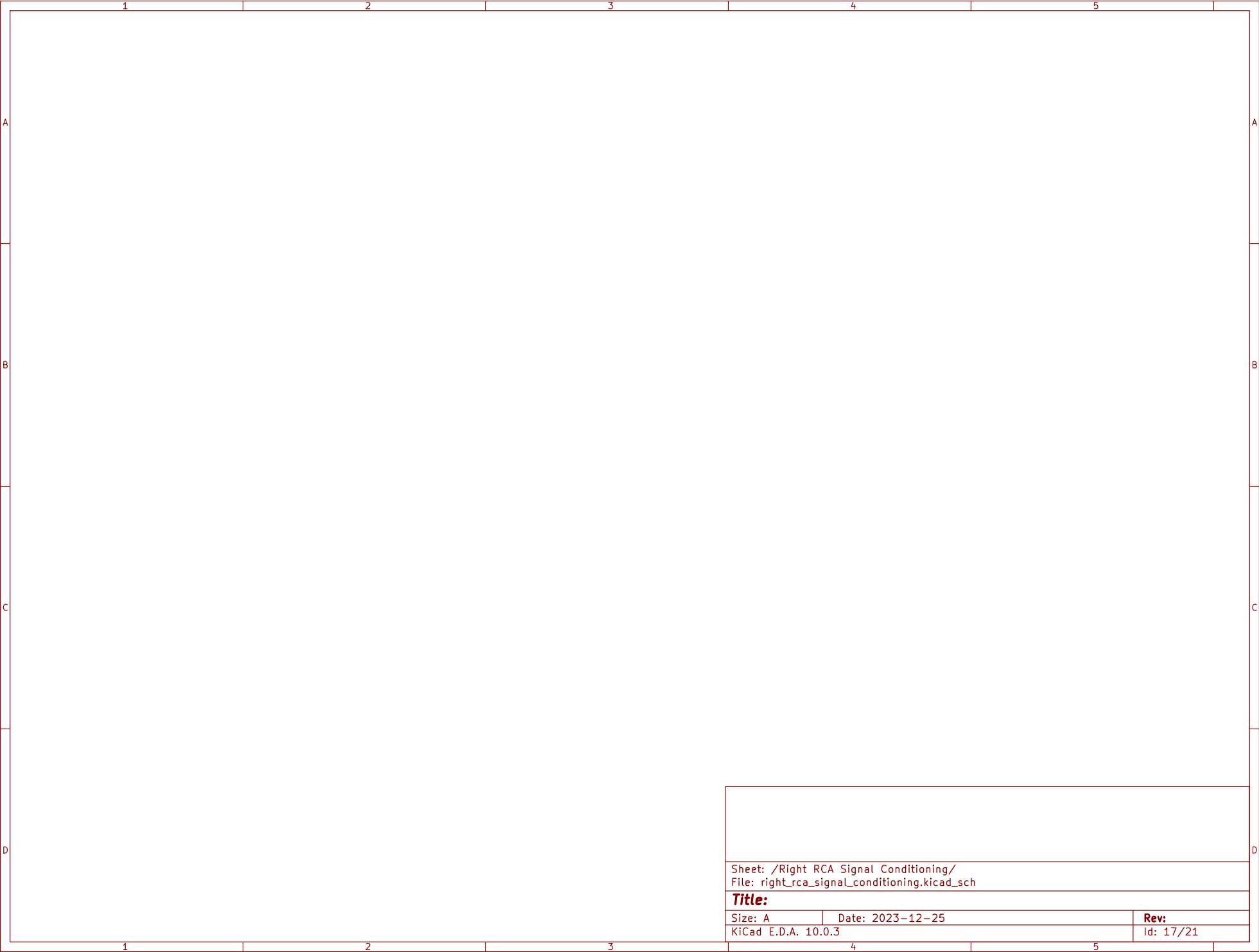
Date: 2023-12-25

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KiCad E.D.A. 10.0.3

Id: 16/21

|                                                                                       |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
|---------------------------------------------------------------------------------------|--|--|--|--|------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|------|--|--|--|--|
| Sheet: /Left RCA Signal Conditioning/<br>File: left_rca_signal_conditioning.kicad_sch |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
| Title:                                                                                |  |  |  |  |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |
| Size: A                                                                               |  |  |  |  | Date: 2023-12-25 |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Rev: |  |  |  |  |
| KiCad E.D.A. 10.0.3                                                                   |  |  |  |  | Id: 16/21        |  |  |  |  |  |  |  |  |  |  |  |  |  |  |      |  |  |  |  |



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| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /Left Speaker Signal Conditioning/  
File: left\_speaker\_signal\_conditoning.kicad\_sch

Title:

Size: A

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KiCad E.D.A. 10.0.3

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|----------------------------------------------------------------------------------------------|------------------|------|
| Sheet: /Left Speaker Signal Conditioning/<br>File: left_speaker_signal_conditoning.kicad_sch |                  |      |
| Title:                                                                                       |                  |      |
| Size: A                                                                                      | Date: 2023-12-25 | Rev: |
| KiCad E.D.A. 10.0.3                                                                          | Id: 18/21        |      |

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| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

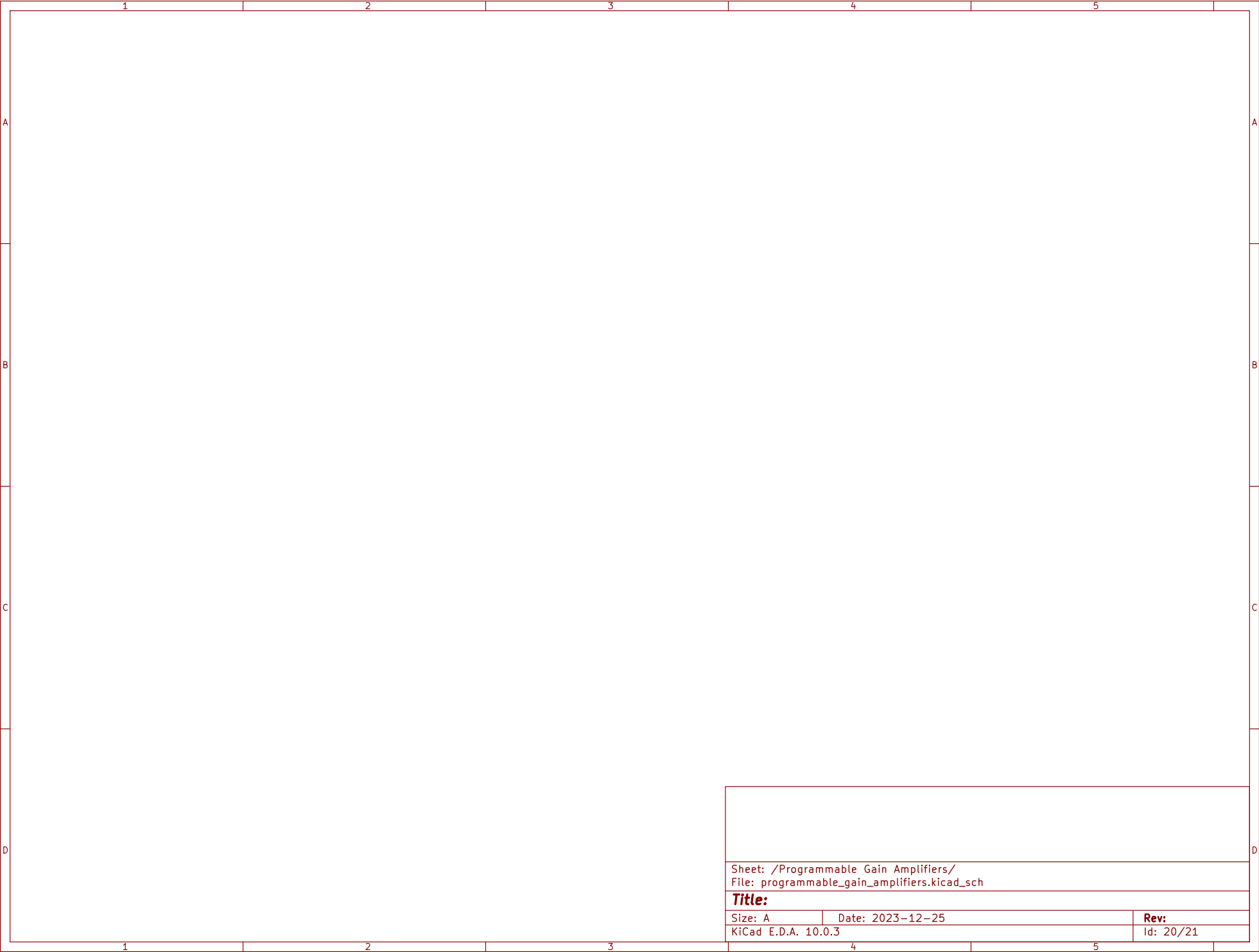
Sheet: /Right Speaker Signal Conditoning/  
File: right\_speaker\_signal\_conditoning.kicad\_sch

Title:

Size: ADate: 2023-12-25

KiCad E.D.A. 10.0.3Id: 19/21

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| Sheet: /Right Speaker Signal Conditoning/<br>File: right_speaker_signal_conditoning.kicad_sch |                  |      |
| <b>Title:</b>                                                                                 |                  |      |
| Size: A                                                                                       | Date: 2023-12-25 | Rev: |
| KiCad E.D.A. 10.0.3                                                                           | Id: 19/21        |      |



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| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |
| A |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | A |
| B |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | B |
| C |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | C |
| D |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  |   |  |  |  |  | D |
| 1 |  |  |  |  | 2 |  |  |  |  | 3 |  |  |  |  | 4 |  |  |  |  | 5 |  |  |  |  |   |

Sheet: /Power Button/  
File: power\_button.kicad\_sch

Title:

Size: A

Date: 2023-12-25

Rev:

KiCad E.D.A. 10.0.3

Id: 21/21

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| Sheet: /Power Button/<br>File: power_button.kicad_sch |                  |           |
| Title:                                                |                  |           |
| Size: A                                               | Date: 2023-12-25 | Rev:      |
| KiCad E.D.A. 10.0.3                                   |                  | Id: 21/21 |